

AI System to Automatically Review and Summarize Research Papers – Milestone 2

Statement of Purpose

The objective of Milestone 2 is to implement an advanced PDF text extraction and analysis module. This phase ensures accurate parsing of downloaded research papers and structured organization of content into sections such as Abstract, Introduction, Methodology, Results, and Conclusion. It also introduces key findings extraction and cross-paper comparison mechanisms.

Scope of Project

This milestone focuses on improving PDF parsing accuracy, organizing extracted content systematically, identifying important research contributions, and performing comparative analysis across multiple papers. The structured data supports efficient summarization and further processing.

Detailed Methodology

- Step 1: Developed advanced PDF text extraction module to handle complex layouts and multi-column formats.
- Step 2: Implemented section-wise parsing logic for Abstract, Introduction, Methodology, Results, and Conclusion.
- Step 3: Structured extracted content into organized formats such as JSON for efficient analysis.
- Step 4: Designed key findings extraction logic using keyword and pattern-based techniques.
- Step 5: Implemented cross-paper comparison module to analyze similarities and differences.
- Step 6: Validated and verified extracted data for completeness and correctness.

Week 3–4 Implementation Tasks

- Advanced PDF text extraction implementation.
- Section-wise content organization and structured storage.
- Key findings extraction logic development.
- Cross-paper comparison algorithm implementation.
- Validation and debugging of extracted textual data.

Algorithm Flow Explanation

1. Start -System begins execution.
2. Load Downloaded PDFs.
3. Perform Text Extraction.
4. Parse Section-wise Content.
5. Store Structured Text Data.
6. Extract Key Findings.
7. Perform Cross-Paper Comparison.
8. Validate Extracted Data.
9. Stop – Process completes successfully.